

SP2718F — Data Science in Action

Final Project Report

[Track A /Issuer]

Group Number	Group 1
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Declaration

By submitting this report, all team members listed above acknowledge that they have reviewed its full content and take collective responsibility for it.

Executive Summary

This project investigates card-not-present (CNP) fraud in the IEEE-CIS dataset from the perspective of the card issuer, which must detect fraudulent transactions while limiting disruption to legitimate customers. Using card, address, email, missingness, and match-related evidence, the team developed four complementary pipelines and combined them through a final decision layer. We then compared our results with a XGBoost Tree model executed on raw columns (recall rate 25.06%, FP rate 0.33%).

The final combined system achieved 36.45% fraud recall, 1.04% false-positive rate, and 0.892 of ROC-AUC, with **pipeline 2 (Card–Address and Card–Email relationship XGBoost Pipeline)** providing the strongest overall trade-off. We recommend that the issuer use the system to support step-up authentication or manual review for high-risk transactions, rather than automatic decline, because this approach improves fraud detection while controlling customer friction.

1. Problem Framing

1.1 Stakeholder and Fraud Type

The stakeholder is the card issuer, which must decide whether to approve, challenge, review, or decline a card-not-present transaction. The issuer aims to detect fraudulent use of card details while limiting disruption to legitimate customers.

1.2 Research Questions and Hypotheses

1. Do purchaser–recipient email relationship categories (matched/different/missing) show significantly different fraud rates, and does treating missing as its own category improve fraud detection?

Hypothesis: Matched, different and missing email-domain relationships may carry distinct fraud signals; missing values should therefore remain a separate category.

2. Do summary features counting available, missing and mismatch-like values across M1–M9 identify higher-risk transactions more effectively than using M1–M9 indicators separately?

Hypothesis: Counts of missing, available and failure-like M indicators may capture broader risk patterns more effectively than individual flags.

3. How does fraud rate vary across combinations of card network (card4) and card category (card6), and can these combinations provide useful transaction-time risk signals?

Hypothesis: Some card-network and card-category combinations may have higher fraud rates and provide useful issuer-side signals.

4. Do card–address and card–email relationship features improve fraud detection over the original fields?

Hypothesis: Training-only frequency, rarity, novelty and relationship-breadth features may represent unusual cardholder relationships more effectively than the original encoded fields alone.

1.3 Success Metric

A good result for the issuer is one that detects a meaningful share of fraudulent transactions while keeping disruption to legitimate customers at an acceptable level. The project therefore adopts a **minimum viable target of at least 25% fraud recall with a false-positive rate below 5% (target recall set against the baseline)**. Recall is the primary metric because missed fraud may lead to disputes and chargebacks. The false-positive rate is equally important because incorrectly flagging legitimate transactions may block real customers at checkout, increasing customer friction. Precision and AUC are also used, but only as a supporting measure of overall ranking quality.

2. Data and Exploratory Analysis

2.1 Dataset Summary

The analysis used all 413,378 transactions in the training data, corresponding to 70% of the full IEEE-CIS dataset sorted in ascending order of TransactionDT [1]. The overall fraud rate was approximately 3.5%, confirming substantial class imbalance. The original joined dataset contained 434 variables, but the project focused on 35 selected columns (see Appendix A), including transaction and product fields, card and address variables, email domains, M-indicators, selected masked C, D, V and id features, and device fields. There are some aspects to be careful with: TransactionDT is an elapsed-time offset with an undisclosed reference point, rather than a real timestamp. Card and address fields are anonymised codes, while C, D and V variables have masked meanings and are therefore interpreted only as empirical risk signals. Identity fields cover only a minority of transactions, so their absence may reflect data coverage rather than suspicious behaviour.

2.2 Top 3 EDA Findings

Finding 1: M-indicators were not simple “mismatch = fraud” signals

#	RowID	Variable [T: String]	status [T: String]	fraud rate [.00 Number (Float)]
1	Row0	M1	missing	0.048
2	Row1	M1	F	0
3	Row2	M1	T	0.02
4	Row3	M5	missing	0.037
5	Row4	M5	F	0.027
6	Row5	M5	T	0.04
7	Row6	M2	missing	0.048
8	Row7	M2	F	0.034
9	Row8	M2	T	0.018

Diagram 1 - fraud rates for M1, M2 and M5 at different status

Missing M1 and M2 values had fraud rates of approximately **4.8%**, while their matched or mismatch groups were generally lower. M5 showed a different pattern, with its matched group reaching approximately **4.0%**. These results show that the M-fields should not be interpreted through one universal rule and motivated by separate missing, failure and availability counts.

Finding 2: Email relationships mattered, but not in the expected direction

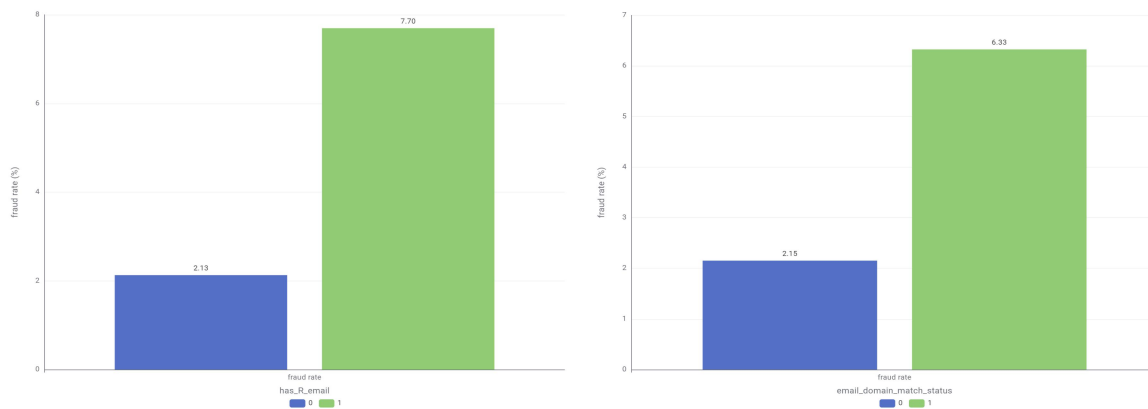


Diagram 2 - fraud rates for has_R_email = 0 & 1 | Diagram 3 - fraud rates for email_domain_match_status = 0 & 1

Transactions with a non-missing recipient email (has_R_email = 1) had a fraud rate of approximately **7.7%**, compared with **2.1%** when the recipient email was missing. Matched purchaser and recipient domains (email_domain_match_status = 1) also had a higher fraud rate, approximately **6.3%**, than different domains at **2.2%**. This refuted the simple assumption that email mismatch is necessarily riskier and supported treating matched, different and missing relationships as separate categories.

Finding 3: Available identity or device evidence was associated with higher fraud

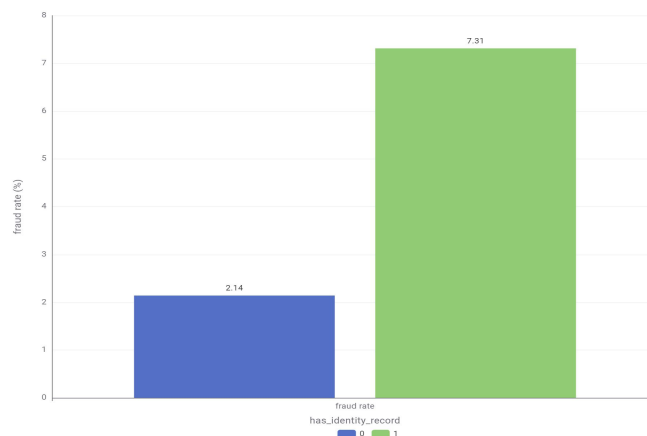


Diagram 4 - fraud rates for has_identity_record = 0 & 1

Transactions with at least one of id_01, id_05 or DeviceType available (has_identity_record = 1) had a fraud rate of approximately **7.3%**; similar rates were observed when each field was examined separately. This led to availability flags and an overall identity-record indicator. However, because identity coverage is incomplete, the result should be treated as an association with a different transaction population rather than evidence that identity availability causes fraud.

3. Data Preparation and Feature Engineering

3.1 Missing Value Strategy

Column(s)	Approx. missing	Likely reason	Treatment	Rationale
card4, card6	<0.01%	Card metadata unavailable for a very small number of records	Impute with mode	Missingness was too rare to justify a separate flag.
addr1, addr2	4.52%	Address code not captured or not mapped	Impute with -1	Preserves the rows while treating missingness as an unknown anonymised address category.
P_emaildomain	14.81%	Purchaser email not collected or unavailable	Fill with "missing"	Email-domain categories may carry fraud information, so missingness was retained explicitly.
R_emaildomain	62.82%	Often structurally absent because no recipient email exists	Create has_Remail flag and retain missing as its own state	Avoids interpreting no-recipient transactions as ordinary email mismatches.
dist1	52.58%	Required underlying records were unavailable	Create has_dist1 flag	Presence or absence was considered more reliable than imputing many artificial distance values.
D4, V34	37.68%, 17.53%	Inputs or reference history needed for these masked fields were unavailable	Median imputation	Median imputation limits the influence of skew and extreme values while retaining the transactions.
id_01, id_05	55.46%, 56.73%	Identity data were only available for some transactions	Create availability flags	Missing identity evidence may distinguish transaction channels and was therefore preserved as a signal.
DeviceType	56.05%	No linked identity/device record or device type not collected	Fill with "missing"	Its low number of categories allowed missingness to be represented directly.
DeviceInfo	High missingness	Device-level text not collected or no identity record available	Create an availability flag	This avoids creating a large, sparse categorical variable while retaining device-evidence availability.
M1–M9 (except M4)	Varies by field	Match check was unavailable or not applicable	Encode T and F, while retaining missing as a separate state	EDA showed that missing, T, and F patterns had different fraud rates; missing values were also used to construct m_missing_count and related summaries.

3.2 Engineered Features

Feature(s)	Definition	Rationale
m_failure_count	Number of M1–M9 fields equal to F	Summarises the amount of mismatch-like evidence without assuming that any single M-field proves fraud.
m_missing_count	Number of missing values across M1–M9	EDA showed that missing M-patterns could be more informative than mismatches alone.
M-pattern flags	Separate missing and failure flags for each M-field; M5_matched_flag for M5 = T	Retains field-specific patterns that may be lost in the summary counts.
Email_domain_match_status	Indicates whether P_emaildomain and R_emaildomain match	Represents purchaser–recipient email consistency. Missing recipient email was retained separately rather than treated as a mismatch.
has_Remail	1 when R_emaildomain is available	Distinguishes transactions with recipient-email evidence from those where no recipient email is recorded.
card_network_funding	Combination of card4 and card6, such as visa_credit or mastercard_debit	Captures joint card-network and credit/debit-like patterns that may have different fraud exposure.
card_addr_freq and card_addr_rare	Training-set frequency of a card–address combination; rare when frequency was below the selected threshold	Tests whether infrequent card–address relationships represent unusual transaction behaviour.
card_addr_unseen	1 when a validation card–address combination was absent from training	Represents relationship novelty without using validation outcomes.
Card/email relationship features	Training-set frequency and novelty measures for card–email combinations	Tests whether new or uncommon card–email relationships add information beyond the raw fields.
Relationship amount deviations	TransactionAmt minus the mean amount for its card–address or card–email group	Identifies transactions whose amount is unusual relative to their associated relationship history.
amt>0.9quantile_by_CD	1 when TransactionAmt exceeds the training-set 90th percentile within its ProductCD	Measures whether an amount is unusually high relative to its product category rather than against one global threshold.
Log-transformed variables	log(TransactionAmt), log(C1+1), log(C9+1) and log(V130+1)	Reduces strong right-skew and limits the influence of extreme values. Masked variables remain interpreted only as numerical risk signals.
Identity availability features	has_id01, has_id05, has_devicetype, and has_identity_record	Retains whether identity or device evidence was available at transaction time.
Missingness-group flags	identity_missing_flag, email_missing_flag, distance_missing_flag, and address_missing_flag	Captures broader absence patterns that may distinguish transaction channels or evidence availability.
Rule-score features	amount_points, card_category_points, evidence_point, and m_points	Converts selected verified signals into an auditable points-based risk score for the transparent pipeline.
total_risk_score	Sum of the retained rule-score components	Produces ordered risk tiers and a threshold-based review decision that can be explained and adjusted by the issuer.

3.3 Leakage Safeguards

The dataset was split chronologically using TransactionDT before model fitting and before calculating any data-derived relationship features. Frequency, rarity, novelty and relationship-breadth statistics for card–address and card–email combinations were computed from the training data only. Validation and test transactions then looked up these fixed training statistics rather than recalculating them; unseen combinations were assigned frequency zero and flagged as novel.

Product-specific amount thresholds, category mappings, missing-value imputations, encodings and normalisation parameters were also fitted on the training data and then applied unchanged to later data. The target isFraud was used only for model training and subsequent evaluation, and was not used to construct transaction-time features, lookup tables or the rule-based score.

4. Multi-Pipeline System

4.1 System Diagram

Layout of the workflow is changed due to the length of the shared part, pipeline 2 (XGBoost Tree) and pipeline 3 (scoring system). For a clearer view (and layout) of these parts, please see in the submitted KNIME files.

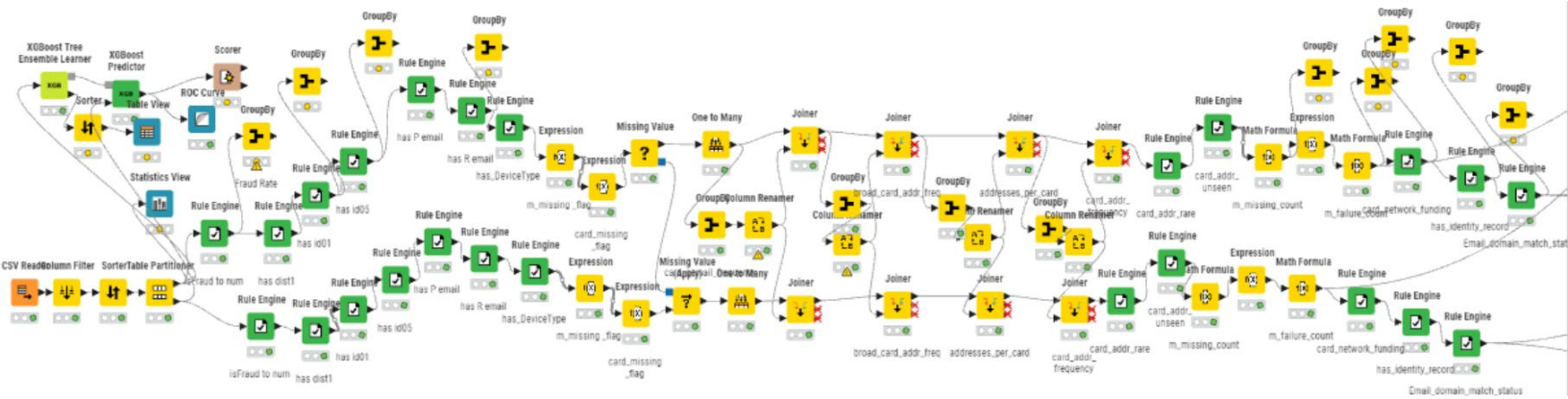


Diagram 5 - Preprocessing & Shared Feature-Engineering Part

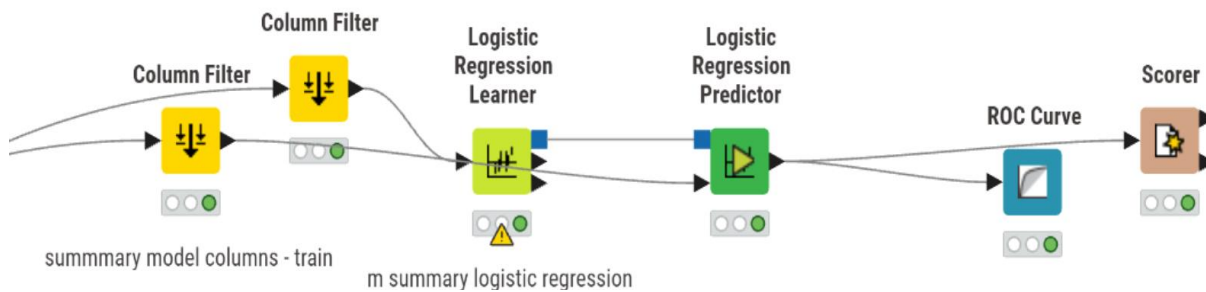


Diagram 6 - Pipeline1: M-Pattern Logistic Regression Pipeline (Chen Jian)

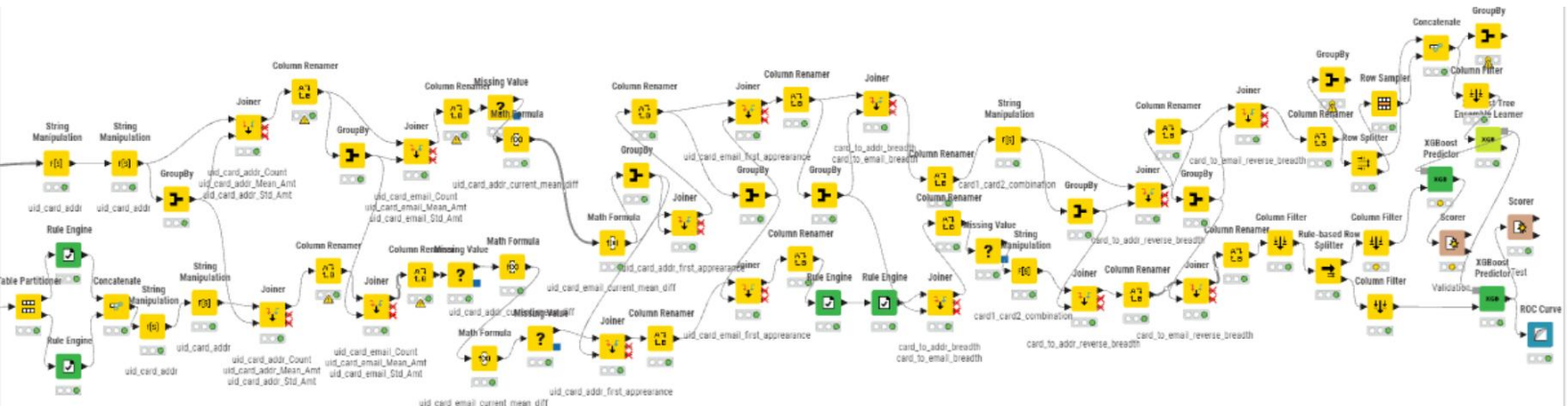


Diagram 7 - Pipeline2: Card–Address and Card–Email XGBoost Pipeline (Yuan Ruotian)

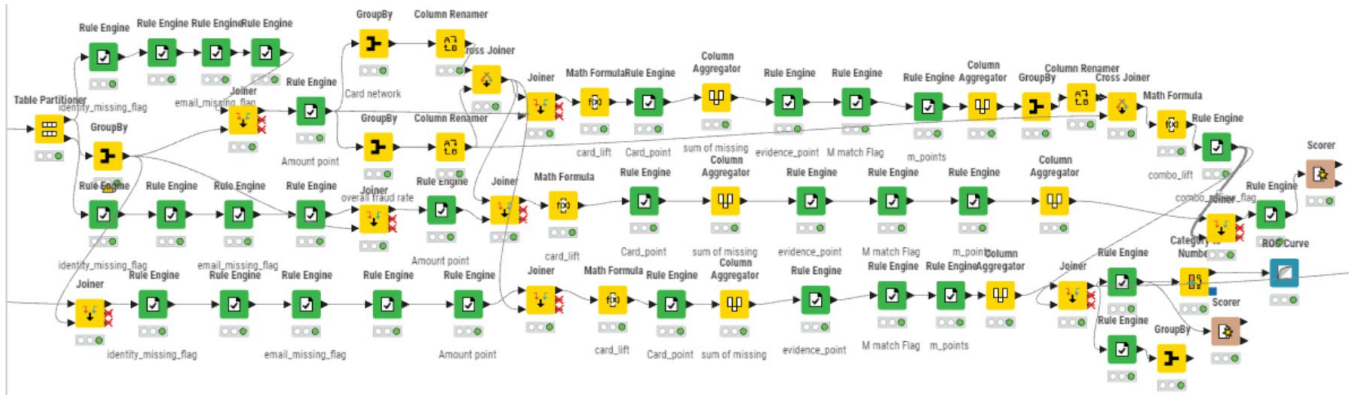


Diagram 8 - Pipeline3: Transparent Rule-Based Risk Score (Zhang Xiaohan)

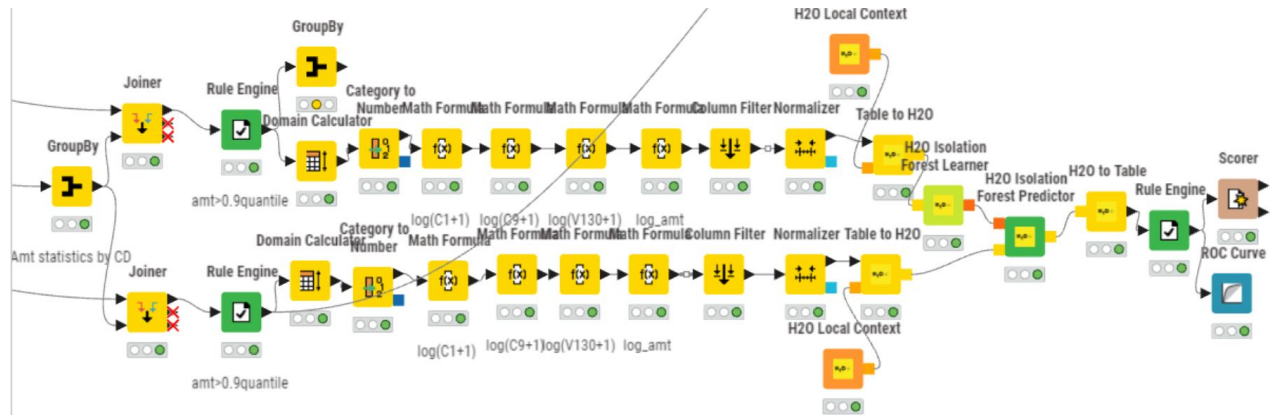


Diagram 9 - Pipeline 4: Isolation Forest (Zhang Yang)

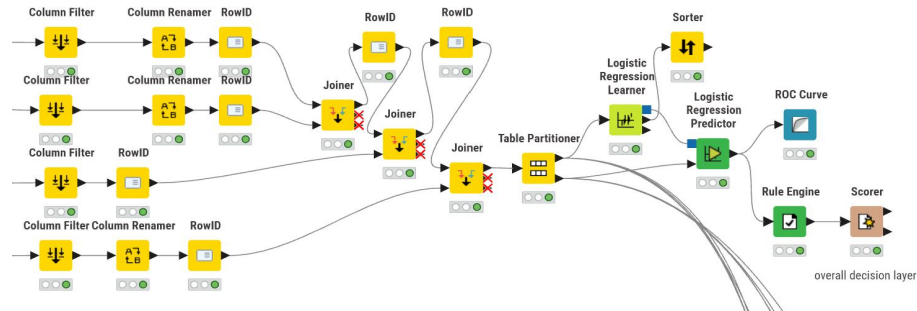


Diagram 10 - Decision Layer: Logistic Regression Stacking Layer

4.2 Individual Pipelines

Pipeline 1: M-Pattern Logistic Regression Pipeline by Chen Jian

An interpretable logistic regression model using M1–M9-related features to test whether match, mismatch, and missing-value patterns can help identify fraudulent transactions [4].

The pipeline uses the original M1–M9 variables, separate counts of missing and failure-like M indicators, combined M-pattern counts, and one-hot encoded M4 categories. It evaluates whether individual M flags or aggregated M-pattern features provide useful fraud signals, using isFraud as the target and a chronological train-validation split based on TransactionDT.

Pipeline 2: Card–Address and Card–Email XGBoost Pipeline by Yuan Ruotian

This pipeline uses an XGBoost classification model to test whether card–address and card–email relationship patterns improve fraud detection beyond isolated card, address, and email variables [2]. It creates interaction identifiers for specific card–address and card–email combinations and converts them into frequency, novelty, transaction-behaviour, and relationship-breadth features. Frequency features measure how often each relationship appears, while historical mean and standard deviation statistics describe its typical transaction amount. Amount-deviation features identify transactions that differ from the relationship's previous behaviour. Novelty

features capture when a relationship first appeared and the time since its first use. Relationship-breadth features measure how many addresses are linked to a card and how many cards are linked to an address, helping detect one-to-many fraud networks. XGBoost was selected because it can capture nonlinear interactions among these engineered features [5]. To address class imbalance, the training data was resampled so fraudulent transactions formed one-third of the training set. A validation set was used for model and threshold tuning, while the test set remained unseen. All relationship statistics were calculated from training data only and applied unchanged to validation and test data, preventing leakage.

Pipeline 3: Transparent Rule-Based Risk Score by *Zhang Xiaohan*

Pipeline 3 converted selected fraud signals into a transparent points-based risk score. Four data-supported components were retained: unusually high transaction amount relative to ProductCD, higher-risk card categories, availability of selected identity or device evidence, and M-pattern signals based on missingness and $M5 = T$. The scoring rules assigned 1.5 points for high amount, up to 1 point for card category, 1.5 points for identity/device evidence, and 1 point for the M-pattern condition. Card-category rules were only used for groups with at least 200 transactions and 10 fraud cases to reduce instability. Relationship rarity was tested but excluded because it did not show higher fraud rates on the larger dataset. The final score ranged from 0 to 4.5 and was grouped into low, medium and high risk. Transactions scoring at least 3 were flagged directly, while borderline cases were flagged only when their exact score combination showed lift of at least 2.0 with sufficient sample size.

Pipeline 4: Isolation Forest by *Zhang Yang*

This pipeline uses H2O Isolation Forest to identify transactions with unusual combinations of transaction, card, address, email, missingness and M-pattern features [2][6]. Unlike the supervised models, it does not learn directly from isFraud; instead, it produces an anomaly score showing how unusual each transaction is.

The model used selected numerical and encoded features, including amount, card and distance fields, masked C, D and V signals, identity-availability flags, email-match status, M-summary features and card-relationship measures. The anomaly score was converted into a fraud prediction using a validation-selected threshold. Its main contribution is to detect unusual cases that may be missed by supervised models, although unusual legitimate transactions may also increase false positives.

4.3 Decision Layer

The four pipeline outputs were combined using a stacked Logistic Regression model:

- M-pattern Logistic Regression score;
- card–address/email XGBoost score;
- transparent rule-based risk score;
- Isolation Forest anomaly score.

The scores were joined by RowID and used as inputs to a Logistic Regression decision layer in KNIME. The model produced a combined fraud probability, which was converted into the final classification using a Rule Engine and the selected probability threshold. Performance was then measured against isFraud using the Scorer node on the held-out evaluation data.

5. Evaluation

5.1 Combined System Performance

5.1.1 Confusion Matrix and ROC Curve

On the held-out set, the combined decision layer produced 666 true positives, 532 false positives, 1,161 false negatives and 50,790 true negatives (see Appendix B).

Actual / Predicted	Legitimate (0)	Fraud (1)
Legitimate (0)	50,790	532
Fraud (1)	1,161	666

This corresponds to **36.45% fraud recall**, a **1.04% false-positive rate**, and **55.59% precision** at the selected operating threshold. The system therefore exceeded both the minimum target of 25% recall with FPR below 5%.

The combined decision layer achieved an **AUC of 0.892** on the held-out set as shown below, indicating strong overall ability to rank fraudulent transactions above legitimate ones.

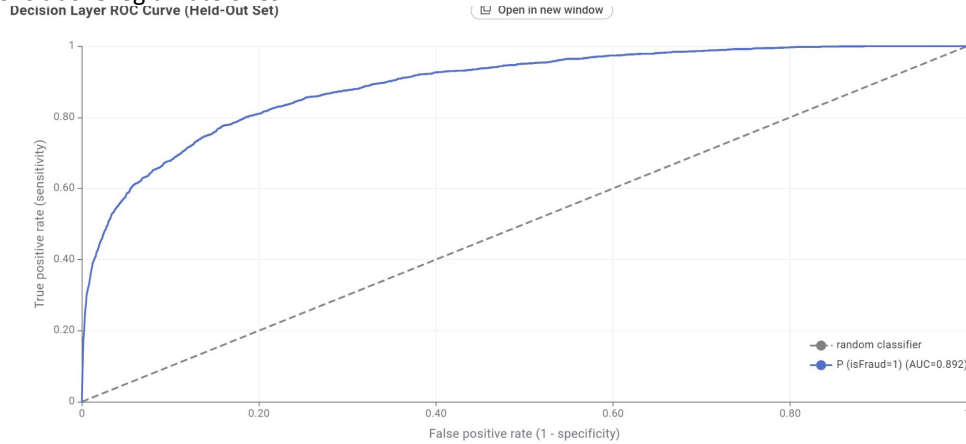


Diagram 10 - ROC-AUC for the final decision layer

5.1.2 Top Features

The XGBoost feature-importance output showed that `addr2`, missing `DeviceType`, `V34`, missing `M4`, and recipient-email availability were the five most influential features. We sorted features by **Gain** because it measures how much each feature improved XGBoost's prediction quality across tree splits [5], making it a useful indicator. These results support the importance of address, missingness, email and match-related evidence in fraud detection. However, gain does not show whether the feature increases or decreases fraud risk, and masked variables such as `V34` cannot be assigned a confirmed business meaning.

Feature	Gain	Treatment
<code>addr2</code>	1791.019	An anonymised address-related field provided the strongest split improvement.
<code>DeviceType=missing</code>	677.058	Missing device information was an important transaction-context signal.
<code>V34</code>	577.276	This masked numerical Vesta feature contributed strongly, but its business meaning is unknown.
<code>M4=missing</code>	419.476	Missingness in the <code>M4</code> match indicator was an important risk signal.
<code>has_Remail</code>	402.149	Availability of the recipient email field helped distinguish transaction risk.

Also, the final decision layer used only the four pipeline scores as inputs. Therefore, its feature interpretation is based on the Logistic Regression coefficients assigned to those scores rather than on raw transaction variables. These coefficients show the relative direction and strength of each pipeline's contribution to the combined fraud probability.

Pipeline Score	Coefficient	Interpretation
<code>xgb_score</code>	4.855	Main predictive signal; strongly increases the final fraud probability when XGBoost identifies high risk.
<code>iso_forest_score</code>	4.546	Strong complementary signal; helps the system recognise unusual transactions not fully captured by supervised models.
<code>lr_score</code>	0.372	Adds additional evidence from M-patterns and improves fraud coverage.
<code>total_risk_score</code>	-0.088	Almost no impact on the final decision. However, this model is still kept as it adds transparency and explainability in business terms.

constant	-6.99	Baseline intercept of the Logistic Regression model
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5.2 Marginal Contribution of Each Pipeline

Each pipeline score was removed in turn and the decision layer was rerun on the same held-out set. The confusion matrices see **Appendix D**, while the PR-AUC calculations and KNIME outputs see **Appendix F**.

Decision-layer version	Recall	False-positive rate	PR-AUC	Main implication
All four pipelines	36.45%	1.04%	0.425	The result of the final combined system.
Without M-pattern Logistic Regression	33.77%	0.92%	0.423	The LR model provides a small but useful increase in fraud detection, and is potentially useful complementary evidence based on M-pattern behaviour.
Without XGBoost	5.80%	0.35%	0.135	The decision layer depends heavily on XGBoost for identifying fraudulent transactions.
Without rule-based score	38.92%	1.23%	0.423	Retain explainability and policy transparency. Provides modest false positive rate control.
Without Isolation Forest	37.71%	1.56%	0.403	Isolation Forest is not mainly finding additional fraud. Instead, within the stacking configuration, it appears as a conservative filtering or false-positive-control component.

The ablation results show that the pipelines contributed in different ways. XGBoost supplied the core predictive signal, while Isolation Forest contributed the clearest secondary improvement in PR-AUC and false-positive control. The M-pattern Logistic Regression added modest fraud coverage. The rule-based score had limited effect on predictive metrics, but provided an interpretable risk signal for the wider system.

5.3 Interpretation

For the issuer, the combined system provides a practical balance between fraud detection and customer friction. It detected 666 fraud cases, equivalent to **36.45% recall**, while incorrectly flagging only **532 legitimate transactions**, giving a **1.04% false-positive rate**. Its **55.59% precision** means that slightly more than half of the transactions selected for intervention were confirmed fraud, while an **AUC of 0.892** indicates strong overall fraud-risk ranking.

The system is therefore suitable for targeted **step-up authentication or manual review**, rather than broad automatic decline. XGBoost supplied most of the predictive strength, while the M-pattern model added fraud coverage and Isolation Forest helped reduce false positives. The transparent score added limited measurable performance but remains useful for explaining why a transaction was considered risky.

6. Recommendation to Your Stakeholder

We recommend using the combined system to trigger **step-up authentication or manual review** for transactions flagged as high risk [3], while allowing lower-risk transactions to proceed normally. This recommendation follows from the held-out results: the system achieved **36.45% fraud recall**, a **1.04% false-positive rate**, **55.59% precision**, and **AUC of 0.892**.

The system should not be used as an automatic decline rule because it still misses some fraud and some flagged transactions are legitimate. Instead, it is best used as a targeted intervention layer that helps the issuer catch more fraud while limiting unnecessary customer friction.

7. LLM Usage Summary

7.1 Prompt Log Summary

The prompt log contains 12 selected prompts covering problem framing, data preparation, feature engineering, pipeline design and evaluation. We used DEPTH, chained, data-aware and feedback prompting, supported by actual missingness rates, confusion

matrices and ablation results. The most useful outputs came from prompts with clear stakeholder context, numerical evidence and explicit constraints. Full analysis and evaluation details can be seen in the prompt log document.

7.2 Three Most Useful Prompts

Prompt 1: Building a Stakeholder-Specific Evaluation Framework

What we asked: For our issuer-side CNP fraud question, what metrics should we use to compare the baseline, individual pipelines and final decision layer? Explain which metric should be primary, how false positives represent customer friction, and why accuracy or ROC-AUC alone is insufficient.

What we got: The response recommended evaluating models using fraud recall, precision, false-positive rate, false-positive volume, confusion matrices and AUC under a shared held-out setup. It also explained that comparisons should be made at a common operating condition, such as the same false-positive rate or review volume, because a model should not be considered better simply because it flags many more legitimate transactions.

Why it was useful: This prompt helped translate the issuer's business objective into measurable evaluation criteria. It shaped the project's focus on fraud recall and false-positive rate, supported the use of confusion matrices and ROC-AUC in Section 5, and clarified why high accuracy is misleading in an imbalanced fraud dataset. It also provided the basis for comparing the combined decision layer with the strongest individual pipeline and the baseline.

Prompt 2: Designing a Meaning-Aware Missing-Value Strategy

What we asked: Here is the task and the data dictionary. Please review how the missing values should be addressed for our issuer-side CNP fraud question. Distinguish unavailable evidence from observed mismatch, recommend treatments for each feature group, and avoid unsupported fraud claims.

What we got: The response proposed different treatments according to feature meaning rather than applying one rule to every column. It recommended preserving missing card, address, email and M-fields as explicit states, treating M1–M9 as match, mismatch or missing, using availability flags for identity and device evidence, and imputing genuinely continuous variables such as distance fields using training-set statistics. It also explained how to handle missing values safely in card–address and card–email relationship features.

Why it was useful: This prompt directly shaped the project's missing-value strategy and engineered features. It helped us avoid treating unavailable evidence as a confirmed mismatch or fraud signal, while supporting features such as `m_missing_count`, `has_Remail`, identity-availability flags and relationship-completeness indicators. It also strengthened the leakage safeguards by requiring imputation and relationship statistics to be learned from training data only.

Prompt 3: Ablation Analysis of Pipeline Contributions

What we asked: Identify the main predictive pipeline, the pipeline that mainly controls false positives, and any pipeline with limited measurable contribution. We chose to keep the rule-based score mainly for transparency and explainability, although it also reduces the false-positive rate at the cost of some recall.

What we got: The response compared the full decision layer with versions that removed one pipeline score at a time. It identified XGBoost as the main predictive component, Isolation Forest as the strongest false-positive-control component, the M-pattern Logistic Regression as a modest source of additional recall, and the rule-based score as mainly contributing transparency, auditability and limited false-positive control.

Why it was useful: This prompt directly supported the report's marginal-contribution analysis by translating the ablation results into clear roles for each pipeline. It also prevented us from overstating the contribution of weaker pipelines and helped justify why a multi-pipeline system was retained even when not every component produced large predictive gains.

8. Reflection

8.1 With More Time or Data

With more time, we would test additional probability thresholds and estimate the financial trade-off between fraud prevented, false alerts and customer friction. With more data, we would also validate the system in a later time period and examine whether feature importance and performance remain stable as fraud patterns change.

8.2 Where the LLM Helped and Where It Did Not

The LLM was useful for structuring the problem, proposing feature treatments, checking metric calculations and translating technical results into issuer-focused language. However, some initial suggestions were too general or based on untested assumptions. The team therefore verified all recommendations using the data, KNIME outputs and course materials, and corrected the LLM when actual results contradicted its expectations.

References

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- [2] KNIME AG. *KNIME Analytics Platform Documentation*. KNIME, accessed July 2026.
- [3] EMVCo. *EMV® 3-D Secure*. EMVCo, accessed July 2026.
- [4] Hosmer, D. W., Lemeshow, S., and Sturdivant, R. X. *Applied Logistic Regression*. 3rd ed., Wiley, 2013.
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- [6] Liu, F. T., Ting, K. M., and Zhou, Z.-H. “Isolation Forest.” *2008 IEEE International Conference on Data Mining*, pp. 413–422, 2008. doi:10.1109/ICDM.2008.17.

Appendix

Appendix A: Selected Variables for general preprocessing and feature engineering

Includes	
☐	TransactionID
☒	isFraud
☐	TransactionDT
☐	TransactionAmt
☒	ProductCD
☐	card1
☐	card2
☐	card3
☒	card4
☐	card5
☒	card6
☐	addr1
☐	addr2
☐	dist1
☐	dist2
☒	P_emaildomain
☒	R_emaildomain
☐	C1
☐	C9
☐	D1
☐	D4
☒	M1
☒	M2
☒	M3
☒	M4
☒	M5
☒	M6
☒	M7
☒	M8
☒	M9
☐	V34
☐	V130
☐	id_01
☐	id_05
☒	DeviceType

Appendix B: Baseline confusion matrix

RowID 0 ☐ Number (Integer)		1 ☒ Number (Integer)	
0	170470		567
1	4590		1535

Appendix C: Individual pipeline confusion matrices

Pipeline 1

RowID	0 Number (Integer)	1 Number (Integer)
0	160223	10814
1	4763	1362

Pipeline 2

RowID	0 Number (Integer)	1 Number (Integer)
0	154786	16251
1	1998	4127

Pipeline 3

2 | Columns: 2

Table  Statistics 

#	RowID	0 Number (Integer)	1 Number (Integer)
1	0	158795	12242
2	1	4442	1683

Pipeline 4

RowID	0 Number (Integer)	1 Number (Integer)
0	166720	4317
1	5579	546

Appendix D: Decision layer confusion matrices

All four pipelines

RowID	0 Number (Integer)	1 Number (Integer)
0	50512	810
1	1066	761

Without M-pattern Logistic Regression

RowID	0 Number (Integer)	1 Number (Integer)
0	50564	758
1	1080	747

Without XGBoost

RowID	0 Number (Integer)	1 Number (Integer)
0	51242	80
1	1764	63

Without rule-based score

RowID	0 Number (Integer)	1 Number (Integer)
0	50581	741
1	1085	742

Without Isolation Forest

RowID	0 Number (Integer)	1 Number (Integer)
0	50424	898
1	1071	756

Appendix F: Precision–Recall

PR_AUC baseline

RowID	Metric ↓ String	Value Number (Float)
Row0	PR-AUC (trapezoidal)	0.409
Row1	Average Precision	0.409
Row2	Fraud-rate baseline	0.035
Row3	Rows used	177,162

PR_AUC: 1

RowID	Metric String	Value Number (Float)
Row0	PR-AUC (trapezoidal)	0.166
Row1	Average Precision	0.069
Row2	Fraud-rate baseline	0.035
Row3	Rows used	177,162

PR_AUC: 2

R... ↓	Metric String	Value Number (Float)
Row0	PR-AUC (trapezoidal)	0.432
Row1	Average Precision	0.432
Row2	Fraud-rate baseline	0.035
Row3	Rows used	177,162

PR_AUC: 3

RowID	Metric String	Value Number (Float)
Row0	PR-AUC (trapezoidal)	0.21
Row1	Average Precision	0.058
Row2	Fraud-rate baseline	0.035
Row3	Rows used	177,162

PR_AUC: 4

RowID	Metric String	Value Number (Float)
Row0	PR-AUC (trapezoidal)	0.116
Row1	Average Precision	0.041
Row2	Fraud-rate baseline	0.035
Row3	Rows used	177,162

PR_AUC: decision layer

RowID	Metric   String	Value  Number (Float)
Row0	PR-AUC (trapezoidal)	0.425
Row1	Average Precision	0.425
Row2	Fraud-rate baseline	0.034
Row3	Rows used	53,149

PR_AUC: drop logistic decision layer

RowID	Metric   String	Value  Number (Float)
Row0	PR-AUC (trapezoidal)	0.423
Row1	Average Precision	0.424
Row2	Fraud-rate baseline	0.034
Row3	Rows used	53,149

PR_AUC: drop XGboost decision layer

RowID	Metric   String	Value  Number (Float)
Row0	PR-AUC (trapezoidal)	0.135
Row1	Average Precision	0.135
Row2	Fraud-rate baseline	0.034
Row3	Rows used	53,149

PR_AUC: drop risk Scoring decision layer

RowID	Metric   String	Value  Number (Float)
Row0	PR-AUC (trapezoidal)	0.423
Row1	Average Precision	0.423
Row2	Fraud-rate baseline	0.034
Row3	Rows used	53,149

PR_AUC: drop Isolation decision layer

RowID	Metric 	Value
	 String	
Row0	PR-AUC (trapezoidal)	0.403
Row1	Average Precision	0.404
Row2	Fraud-rate baseline	0.034
Row3	Rows used	53,149